



**BEng, BSc and MEng Degrees 2023–24**

**Open Assessment**

**Department** Computer Science

**Module** Intelligent Systems: Machine Learning and Optimisation (IMLO)

**Title** Individual Project

**Issued** Thursday 25 04 2024

**Submission due** 12 noon, Thursday 23 05 2024

**Feedback and Marks due** Thursday 20 06 2024

**Instructions:**

Answers must not exceed 2 A4 pages in total, using the IEEE Journal Paper Template with its default font and margins. This limit includes any diagrams, tables, and so on. Up to one separate page for only references (bibliography) is allowed. Excess pages will not be marked. Your report must address all sections outlined in the 'Report' section to gain full marks. Your report must be formatted as a single PDF file.

Your assessment should be submitted as a single zip file, which includes your: report, code, trained model, instructions for building your virtual environment, instructions for running your code, and any further requested information.

All students should submit their answers through the electronic submission system: the Department's Teaching Portal by 12 noon, Thursday 23 05 2024. An assessment submitted after this deadline will be marked initially as if it had been handed in on time, but the Board of Examiners will normally apply a lateness penalty.

Your attention is drawn to the section about Academic Misconduct in your Departmental Handbook.

Any queries you may have on this assessment should be posted on the Discussion Board on the Virtual Learning Environment (VLE) page for Intelligent Systems: Machine Learning and Optimisation (IMLO) in the appropriate discussion area. **No questions will be answered after 16 05 2024.**

### Note on Academic Integrity

This is an open assessment, and you are permitted to refer to written and online materials to aid you in your answers. However, you must ensure that the work you submit is entirely your own, and for the whole time the assessment is live, you must not:

- communicate with other students on the topic of this assessment.
- seek advice or contribution from any other third party, including proofreaders, friends, or family members.

We expect, and trust, that all our students will seek to maintain the integrity of the assessment, and of their award, by ensuring that these instructions are strictly followed. Where evidence of academic misconduct is evident, this will be addressed in line with the Academic Misconduct Policy and, if proven, be penalised in line with the appropriate penalty table. Given the nature of these assessments, any collusion identified will normally be treated as cheating/breach of assessment regulations and penalised using the appropriate penalty table (see AM3.3 of the Academic Misconduct Policy).

Please note: The publication of your assessment submission online or elsewhere, or the sharing of these documents with others, is **NOT PERMITTED** until all students have completed the module assessment and reassessment (this will include any students with extensions to their reassessment date). We will notify you through the VLE at such time.

## 1 Scenario & Setting

The scenario for this assessment is the classification of images into specific classes. In machine learning, this is called *image classification*. The aim is to train a machine learning model from an image dataset, that can then, given a 2D input image, classify the content of that image into a set of pre-defined categories. The task might be to classify generic objects into broad categories, for example categories like 'airplane', 'cat' or 'horse', or more specific tasks like classifying digits.

In this assessment, you will construct and train a neural network classifier for the task of flower species categorisation, using the *flowers-102* dataset, collected and prepared by researchers from the University of Oxford. It is one of the well-known datasets for flower classification and contains a predefined set of training and test images.

This assessment requires you to construct a neural network classifier, that will take an image as its input, and predict a specific class as its output. Your task is to construct a suitable (deep) neural network architecture, train it on the flowers-102 training set, and then evaluate its classification accuracy on the official flowers-102 test set. You will then prepare a report in the style of a short academic research paper, presenting your method and results.

### 1.1 Software Framework

This assignment is to be done in Python 3, and you have to use one of the most popular machine learning libraries/frameworks, PyTorch<sup>1</sup>. You have to use the most recent version. For PyTorch, this is either PyTorch 2.2.2 (or newer).

### 1.2 Dataset

The dataset we'll use is the Oxford 102 Category Flower Dataset (*flowers-102*). The details and data download are given on its website: <https://www.robots.ox.ac.uk/~vgg/data/flowers/102/>. However, instead of downloading the data manually we strongly recommend to use the dataset loaders for in PyTorch (<https://pytorch.org/vision/stable/generated/torchvision.datasets.Flowers102.html>) to directly load the datasets into your environment, including the labels and train/validation/test splits.

You are not allowed to use images from any other dataset to train your network, and you have to use the official split into train, validation and test set used by original flowers-102 dataset for your experiments. The training set is to be used to train the model, the validation set is to be used to evaluate the model performance during the training and for hyper-parameter tuning, and the test set is to be used only for the final evaluation.

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<sup>1</sup><https://pytorch.org/>

You may use data augmentation techniques, as long as the images are augmented from the flowers-102 training set, and no other external images or knowledge is used.

### 1.3 Neural Network Architecture

You will have to construct a neural network classifier, that will take an image as its input, and predict a specific class as its output. You will have to use appropriate layers and loss function, and train your network using an optimiser on the dataset.

You may take inspiration from existing network architectures, but you need to design/write the architecture yourself, in your own code, using PyTorch. Using existing models or existing model code from online sources (e.g. GitHub) is not allowed. If you take inspiration from existing code, you need to make it sufficiently clear that it is your own code, and not copy and pasted from the online source.

You need to train your own network from scratch, i.e. using a pretrained network or pretrained weights is not allowed.

### 1.4 Hardware Resources

To ensure fairness and equal opportunity for all, and to not make the assessment about who has the most computing power available, your network **must train within 12 hours**, on one GPU and/or CPU (it can be any model). If you don't have a suitable computer, you can request access to a departmental compute server via [itsupport@york.ac.uk](mailto:itsupport@york.ac.uk), or alternatively to a lab GPU via the Hardware Team. But note that it may likely take a few days from the day of request until access is granted, so plan for that.

### 1.5 Performance Evaluation

After any experimentation and testing, your report must present a final network for the classification task that can be used by others to classify new data. You must use the official flowers-102 test set and report the accuracy on the test set in your report.

## 2 Submission

Your final project mark will be made up of two components: A mark for a report (weighted at 60%), and the reported classification performance of your network (weighted at 40%). Each of the two components is described in the next two subsections.

## 2.1 Report [60% (total 60 marks)]

The report will present an overview to your method and results. You must evaluate thoroughly the performance of the network (in terms of accuracy). You must provide justification for the choices you make.

You are required to use the two-column *IEEE Journal Paper Template*. It is recommended that you use Overleaf<sup>2</sup>, an online LaTeX editor. The IEEE template is available directly in Overleaf via this link: <https://www.overleaf.com/latex/templates/ieee-journal-paper-template/jbbbdkztwxrd>, by clicking “Open as Template” on that website.

If you are creating a new Overleaf account, log in with or link your University of York email address, so that you can benefit from the University’s subscription, which allows to share documents with an unlimited number of participants.

Your report must be **no more than two A4 pages in length**, including any diagrams or tables. A third page for only references (bibliography) is allowed. At the top, it must indicate your exam number: In the IEEE LaTeX template, find the `\author{...}` command, and put your exam number inside. Your report must contain the following sections, in the order given below, and material outside the six prescribed sections will not be marked.

1. [6 marks] Abstract

*This section should give a brief overview of the highlights of the report: What the task is, what you’ve done, how, and what you’ve achieved. Each of these should be around one sentence.*

***The last sentence of the abstract should clearly state your achieved classification accuracy on the flowers-102 test set.***

2. [6 marks] Introduction

*This section should give a brief introduction to what the task is (in your own words), with a few references to the wider field of image classification, how other people have been tackling the task historically and currently, why it is an important machine learning problem, and then lead into how you are tackling the problem.*

3. [12 marks] Method

*This section should introduce your approach: The network you’ve chosen to create, its layers, and the loss function, and how this works to train a model that learns to classify images. You should clearly describe and justify the type of architecture and/or layers that you’ve used. You can go into the full details of your approach here - for example, you could include a formula of the loss function.*

4. [6 marks] A diagram of your network architecture

*Include a figure of the network that you’ve created to achieve your final result. The figure should exactly depict your network in terms of the layers used and their dimensions.*

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<sup>2</sup><https://www.overleaf.com>

*PyTorch includes functions to visualise the graph of a network, but you may likely need to post-process that output to make a nice diagram out of it. Another option would be to draw the figure with a tool such as PowerPoint.*

5. [12 marks] Results & Evaluation

*This section should describe your experiments, the evaluation metrics that you've used, your results, and draw initial conclusions from your results. If you've run multiple experiments, you could include a table with the results of the various experiments you ran. Your experiments should be described in a way such that they are fully reproducible, for example, describe (and justify) your chosen optimisation algorithm, and all the parameters and hyperparameters used for training. You should include here your classification test accuracy, and the setup of your evaluation to compute that. You also have to include the hardware and environment used for training, and how long it took to train your network.*

6. [6 marks] Conclusion & Further work

*This section should draw conclusions from the whole project. Did you achieve a good performance? Is your architecture a good choice? What went well, what did not go very well? This section should also identify at least one item for further work.*

Additional marks will be given for:

- [6 marks] Appropriate and correct use of the IEEE double-column template.
- [6 marks] References to appropriate literature. Your text should show that you have understood the literature you have selected.

The 6 marks for each section (or 12 marks for Method and Results & Evaluation) are awarded using stepped marking<sup>3</sup> according to the following scheme:

- 0 marks: Not present or not meeting the minimum standard
- 2 marks (4 marks for Method/Results): *Low*: Meeting the minimum standard
- 4 marks (8 marks for Method/Results): *Mid*: Meets most of the standards well
- 6 marks (12 marks for Method/Results): *High*: Fully meets, and, in some aspects, exceeds the standards

You are free with regards to the length of the individual sections of the report (as long as the total is below two A4 pages, without references). On the VLE in the "Assessment" section, there is an example report provided with dummy text, to help you gauge the format, structure and length of individual sections.

You need to be able to clearly explain each component of the network that you've built, and justify the overall architecture. If it becomes apparent that you do not understand the architecture

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<sup>3</sup>University of York Guide to Assessment (2020/21), Appendix R

that you've used, it will result in zero marks for the respective section of the report as well as for the classification performance (section 2.2).

The final report must be formatted as a single PDF file.

## **2.2 Classification performance [40% (total 40 marks)]**

The report submission has to clearly state the achieved classification accuracy on the official flowers-102 test set in the abstract and in the *Results & Evaluation* section. Marks are awarded according to the achieved test accuracy of the trained model. It has to be clear from reading your report that you have trained your network only on the flowers-102 training set, and that the test accuracy has been computed only from images in the test set - otherwise you will not be awarded any marks in this section.

Your code needs to be able to reproduce your reported accuracy within a margin of 3% when run/re-run (as there is some randomness involved when training most neural networks).

## **2.3 Code repository & submission**

You **must** submit your code and trained model as part of your submission, including instructions on how to run your code and details for building the virtual environment and packages used (e.g. requirements.txt, environment.yml, etc). This will not be marked, but will be used to validate the reported classification performance. You are expected to use git and either GitHub<sup>4</sup> or GitLab<sup>5</sup> (or a similar platform) throughout your project. You are expected to work on that repository and contribute via *commits*. The markers may ask for access to the repository and its full commit history on a case-by-case basis, and you have to be able to provide such.

**END OF PAPER**

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<sup>4</sup><https://github.com>

<sup>5</sup><https://gitlab.com>